COVID-19 in China — Regression, Classification and Clustering

Analysing daily COVID-19 case trajectories in Chinese provinces through sliding-window regression, logistic classification, and k-means clustering.

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Overview

This project applies three families of machine-learning technique to daily confirmed COVID-19 case counts across Chinese provinces. A sliding-window approach converts Beijing's time series into a tabular regression and classification dataset. A separate clustering study tests whether geographic proximity explains province-level case similarity. The analysis reveals that outbreak magnitude, not physical distance, drives the structure of the data.

Goals

Demonstrate that standard regression, classification, and clustering methods can extract meaningful structure from a real epidemiological time series, and evaluate the limits of each technique on a small, imbalanced dataset.

Objectives

- Build a six-day sliding-window dataset from Beijing's daily case series and generate regression and classification targets.
- Fit linear regression and LASSO models to predict the sixth day from the previous five, and analyse how LASSO regularisation strength affects test MSE.
- Label each window by next-day direction (goup / godown / stable) and sweep logistic regression over L1/L2 penalty and C hyperparameters to study accuracy.
- Cluster all provinces two ways with k-means — by case series and by geographic coordinates — and compare partitions using the adjusted Rand index.
- Draw practical conclusions about the predictability of daily case direction and the relationship between geographic distance and outbreak similarity.

Approach

Data preparation uses a fixed-width sliding window to create supervised datasets from a univariate time series. Regression models are evaluated on held-out test MSE; the LASSO alpha sweep shows how coefficient shrinkage degrades predictive power. For classification, features are standardised before fitting logistic regression, and a grid over penalty type and C provides a structured hyperparameter analysis. Clustering uses k-means on two feature representations — normalised daily counts and latitude/longitude coordinates — with the adjusted Rand index quantifying agreement between the two partitions.

Outcomes

- Linear regression and LASSO both achieved low MSE on the Beijing regression dataset; LASSO MSE rose and plateaued as alpha increased beyond roughly 100.

- Classification accuracy was modest due to the small dataset size (67 windows) and class imbalance; moderate to large C values gave the best results for both L1 and L2 penalties.
- The adjusted Rand index between the geographic and case-based clusterings was near zero, showing the two partitions largely disagree.
- Hubei separated as its own cluster in the case-based grouping due to its outsized outbreak magnitude, while most other provinces clustered together.
- Geographic proximity is a poor predictor of COVID-19 case similarity; outbreak size is the dominant factor.